

FPGA-Based Real-Time ECG Monitoring and QRS Detection System

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Abstract—Electrocardiogram (ECG) monitoring is a vital tool in healthsector, so doctors can analyse how the heart is functioning and quickly spot potential issues. An ECG signal is made up of several key parts—the P wave, QRS complex, and T wave—each providing different insights into heart activity. The QRS complex, in particular, is crucial because it helps doctors measure heart rate and detect abnormal rhythms. Accurately identifying the QRS complex is essential for reliable diagnoses and effective patient care.

However, many traditional ECG analysis systems depend mainly on software, which can have trouble handling continuous, real-time signals smoothly. In this project, we address these challenges by building a real-time ECG monitoring and QRS detection system using simulation and FPGA technology. Our system uses ECG data from the well-established MIT-BIH Arrhythmia Database and processes the signals step by step—including filtering and peak detection—to precisely locate each QRS complex in every heartbeat. The entire system is designed and tested in the Vivado environment to ensure reliability and accuracy.

While our testing so far has been limited to simulations and not run on a physical FPGA board, our promising results suggest strong potential for future real-time ECG monitoring solutions powered by hardware.

Index Terms—MIT-BIH Arrhythmia Database, QRS detection system

I. INTRODUCTION

Electrocardiogram (ECG) monitoring is an essential tool in today's healthcare, letting doctors see how the heart is working and spot any irregularities. The ECG trace shows several key features—like the P wave, QRS complex, and T wave—with the QRS complex being especially important for measuring heart rate and detecting abnormal rhythms. Because of this, accurately finding the QRS complex is vital for diagnosing

heart problems and keeping track of a patient's condition over time.

As the demand for real-time health monitoring grows, traditional software methods for analysing ECG data can find difficult to keep up with the constant flow of information. To tackle this challenge, hardware solutions like Field Programmable Gate Arrays have become a strong alternative. They process data quickly, handle many tasks at once, and use resources efficiently, making them well-suited for this kind of work.

In this project, we propose using an FPGA to monitor ECG signals and detect QRS complexes in real time through simulation. The system works with ECG data from the MIT-BIH Arrhythmia database, using signal processing steps like filtering and peak detection to pinpoint QRS complexes accurately. We developed and tested the design in the Vivado simulation environment, showing how FPGA-based systems can make ECG analysis faster and more dependable.

To improve detection accuracy, the system applies a sequence of processing stages that enhance the ECG signal and reduce noise caused by muscle activity, power line interference, and other disturbances. These preprocessing steps ensure that the QRS complexes are clearly distinguishable from other waveform components. The peak detection algorithm is then used to identify the exact locations of the QRS complexes, which can further be used to calculate most useful parameters such as heartrate and RR intervals.

Another important aspect of this project is efficient hardware design. The FPGA implementation is structured to optimise the use of logic resources while maintaining high performance. By leveraging parallel processing, the system is capable of handling continuous ECG data streams without significant delay, which is crucial for real-time monitoring applications.

Although the current implementation is limited to simulation, it demonstrates the feasibility of deploying such systems on actual FPGA hardware. In the future, this work can be extended to include live ECG signal acquisition and integration with wearable or portable medical devices. This would enable continuous and remote patient monitoring, helping in the early detection of cardiac abnormalities and improving overall healthcare outcomes.

II. LITERATURE REVIEW

Over the years, a variety of methods have been created to analyze ECG signals and detect the QRS complex—a key part of determining heart rate and identifying heart issues. In the early days, most ECG analysis used software techniques like thresholding, differentiation, and digital filtering. One of the known and trusted approaches is the Pan–Tompkins method, which combines filtering, differentiation, squaring, and integration to accurately pick out QRS complexes in ECG data [2]. Later, researchers refined these detection methods to make them even more precise in different situations [3].

As technology in signal processing improved, wavelet transform techniques were introduced to do an even better job of finding important points in ECG signals, especially when the data is noisy [4]. These methods give a more detailed look at both the timing and frequency of ECG signals, which really helps when working with complex heart readings.

To make sure these algorithms actually work well, researchers often use standard datasets like the MIT-BIH Arrhythmia database, which contains a vast range of real-life heart conditions [7]. Tools like PhysioNet have also made it much easier to get access to this kind of biomedical data for research [6]. These resources are crucial for checking how reliable QRS detection systems are in practice.

Lately, machine learning and neural networks have become increasingly important for sorting ECG signals and detecting arrhythmias [8]. These newer methods can be very accurate, but they often need a lot of computing power.

To overcome the limitations of software-based systems, researchers have started turning to hardware solutions like Field Programmable Gate Arrays (FPGAs). FPGA-based ECG systems can handle several tasks at once, making them a great match for real-time heart monitoring [14], [15]. For instance, the CardioSync system has shown just how effective FPGA-based designs can be for analysing ECG signals in real time [1].

There’s also been a strong focus on advanced techniques such as feature extraction, classification, and noise reduction to make heart diagnosis systems even more accurate and dependable [17].

Despite all this progress, it’s still a challenge to balance accuracy, speed, and hardware complexity. Differences in ECG signals from patient to patient and the presence of noise can make detection more difficult. That’s why there’s a constant need to development of systems that are both reliable and efficient for real-time ECG monitoring.

Building on all this earlier work, this project introduces a simulation-based FPGA approach for ECG monitoring and QRS detection.

III. SYSTEM ARCHITECTURE

The system we’re proposing for real-time ECG monitoring and QRS detection is put together like building blocks, with each stage handling its own specific task. In simple terms, the system takes in ECG data, processes it step by step, and produces results like clean signals and clearly marked QRS peaks. By breaking parts, it’s much easier to develop, test, and perfect each section before everything is combined into one working system.

It all starts by bringing in ECG data from the MIT-BIH Arrhythmia database. These signals are already digital and are sampled at a consistent rate, which helps make sure the analysis is reliable. The data is carefully organized and synchronized so that it moves smoothly through each stage of processing.

The first real processing step is cleaning up the ECG signals. Everyday sources of noise—like electrical interference, baseline shifts, or even a patient’s movement—can make the signal messy. To fix this, we use filters that only let through the important parts of the signal and block out the unwanted stuff.

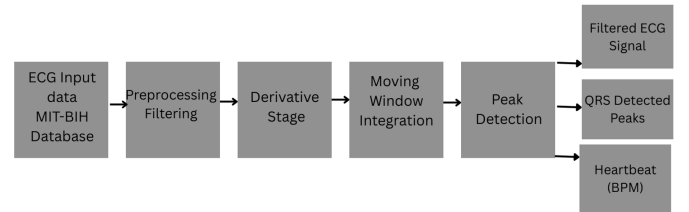


Fig. 1. Blockdiagram

After cleaning, the signal goes through a stage that focuses on changes in the slope of the ECG—this is especially useful for highlighting the QRS complex, which is a key part of the heart’s rhythm. By zeroing in on these rapid changes, it’s easier to spot the QRS complexes and separate them from slower parts like the P and T waves.

Next, we square all the signal values, which simply means making them all positive and making the peaks stand out more. This step makes it easier to spot the most important features and ignores the minor bumps in the signal.

After that, the signal gets smoothed out using a moving window integration. Think of this like running a rolling average over the data to help reveal the width of each QRS complex and to iron out any sudden, noisy spikes. This makes the signal much steadier and easier to analyze in the next steps.

Once the signal is cleaned, highlighted, and smoothed, it goes to the peak detection stage. Here, we use thresholds to pinpoint exactly where the QRS peaks are. These thresholds can even adapt to changing conditions in the signal. Once the peaks are found, we use them to calculate important

measurements, like how far apart the heartbeats are (the RR interval) and the heart rate itself.

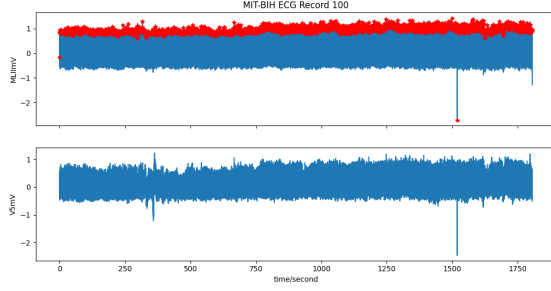


Fig. 2. MIT-BIH ECG Record 100

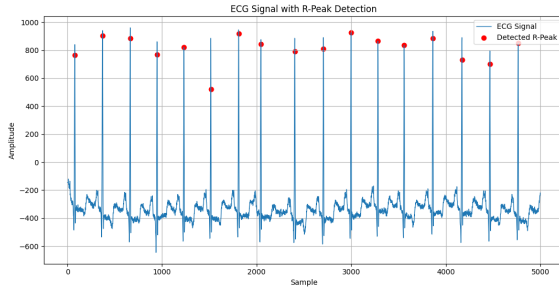


Fig. 3. ECG Signal

Finally, the system gives us outputs such as the filtered ECG signal, the locations of the QRS peaks, RR intervals, and heart rate in beats per minute. These results can be used for deeper analysis or simply displayed visually, making them easier to understand.

The whole system is designed and tested in the Vivado design environment, with each part built to be fast and efficient. This makes the system perfect for real-time use. This design can be loaded onto FPGA hardware, making it possible to use for real-time ECG monitoring in actual healthcare environments.

IV. WORKING PRINCIPLE

Accurate detection of the QRS complex plays a crucial role in electrocardiogram (ECG) analysis, as it represents ventricular depolarization and forms the basis for heart rate calculation, rhythm interpretation, and detection of cardiac abnormalities. However, raw ECG signals are often corrupted by noise and artifacts, making direct detection unreliable.

To address this challenge, a sequence of signal processing techniques is employed, including filtering, slope detection, squaring, moving window integration, and thresholding. Each stage progressively enhances the characteristics of the QRS complex while suppressing unwanted components, thereby improving detection accuracy.

A. Filtering

The first step in the process is filtering, which aims to remove noise such as baseline wander, muscle artifacts, and power-line interference present in the raw ECG signal. Since the QRS complex primarily lies within the frequency range of 5–15 Hz, a band-pass filter is used to retain this range while attenuating other frequencies. The filtering operation can be expressed as:

$$y[n] = \text{Filter}(x[n]) \quad (1)$$

where $x[n]$ represents the input ECG signal and $y[n]$ denotes the filtered output. This step is essential to ensure that noise does not lead to incorrect detections.

B. Slope Detection (Derivative Method)

The QRS complex is characterized by rapid changes in amplitude, resulting in steep slopes compared to other components such as the P and T waves. To highlight these rapid transitions, a derivative operation is applied to the filtered signal. A simple first-order derivative is given by:

$$y[n] = x[n] - x[n - 1] \quad (2)$$

For improved accuracy, a higher-order derivative approximation can be used:

$$y[n] = \frac{1}{8T} [-x[n - 2] - 2x[n - 1] + 2x[n + 1] + x[n + 2]] \quad (3)$$

where T is the sampling interval. This step enhances the slope information and helps distinguish the QRS complex from slower ECG components.

C. Squaring

After slope detection, the signal is squared to make all values positive and to further emphasize larger amplitudes associated with the QRS complex. The squaring operation is defined as:

$$y[n] = [x[n]]^2 \quad (4)$$

This transformation suppresses smaller variations while amplifying significant peaks, making the QRS complex more prominent.

D. Moving Window Integration

The squared signal is then processed using moving window integration, which provides information about both the amplitude and duration of the QRS complex. This step smooths the signal and produces a waveform that reflects the energy content over a short time interval:

$$y[n] = \frac{1}{N} \sum_{i=0}^{N-1} x[n - i] \quad (5)$$

where N is the window length. This operation helps in generating a smoother signal, facilitating more reliable peak detection.

E. Thresholding

The final step involves thresholding, where peaks in the processed signal are compared against a predefined or adaptive threshold to identify QRS complexes:

$$\text{QRS detected if } y[n] > \text{Threshold} \quad (6)$$

Adaptive thresholding techniques are often preferred, as they can adjust to variations in signal amplitude across different subjects or recording conditions. This step ultimately converts the processed signal into discrete detection points, enabling accurate heart rate estimation and rhythm analysis.

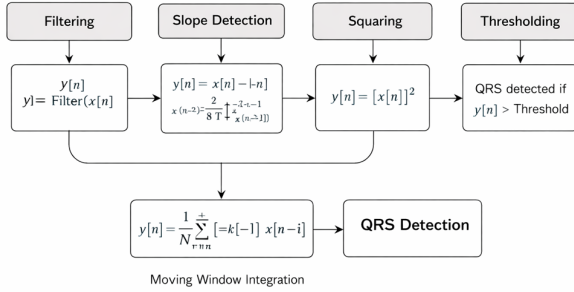


Fig. 4. QRS detection techniques

V. SIMULATION RESULTS AND DISCUSSION

The proposed ECG signal processing system was simulated using a hardware description model, and the internal signals were analyzed to evaluate the effectiveness of each processing stage. The simulation focuses on observing how the input ECG signal is transformed through the pipeline and how accurately the QRS complexes are detected.

A. Waveform Analysis

The input ECG signal (`ecg_in`) initially contains noise and irregular fluctuations. After passing through the filtering stage, the signal (`filtered`) becomes smoother, indicating successful suppression of high-frequency noise and baseline disturbances.

The derivative signal (`deriv`) exhibits sharp transitions corresponding to rapid changes in the ECG waveform. These transitions align with the locations of the QRS complex, confirming that the derivative stage effectively enhances slope information.

Following this, the squared signal (`squared`) shows a significant increase in amplitude for the detected peaks. All signal values become positive, and the difference between QRS peaks and other components becomes more pronounced.

The integrated signal (`integrated`) presents a smooth envelope of the squared waveform. This smoothing effect helps in clearly identifying the duration and position of QRS complexes, even in the presence of minor noise variations.

B. Peak Detection Performance

The peak detection output (`peak`) successfully identifies QRS complexes by generating pulses at locations where the integrated signal exceeds the threshold value. The detection is consistent and aligns with the expected positions of R-peaks in the ECG waveform.

It is observed that the use of thresholding minimizes false detections caused by noise while ensuring that true QRS complexes are accurately captured.

C. Heart Rate Estimation

The heart rate signal (`bpm`) is derived from the time interval between consecutive detected peaks. The computed BPM values remain stable throughout the simulation, indicating reliable detection performance. Minor variations in BPM reflect natural fluctuations in the input signal.

D. System Behavior and Signal Progression

The sequential transformation of signals through the pipeline demonstrates a clear improvement in signal quality at each stage. The progression can be summarized as:

- Noise reduction observed after filtering
- Sharp feature enhancement after derivative operation
- Peak amplification after squaring
- Signal smoothing after integration
- Accurate detection after thresholding

E. Timing and Processing Efficiency

The sample counter (`count`) confirms that ECG samples are processed sequentially without data loss. The pipeline structure ensures continuous data flow, making the system suitable for real-time implementation.

F. Discussion

From the simulation results, it is evident that each processing stage contributes significantly to improving QRS detection accuracy. The combination of derivative, squaring, and integration effectively enhances the signal features required for reliable peak detection.

The system demonstrates robustness against noise and maintains consistent performance, making it suitable for FPGA-based real-time ECG monitoring applications.

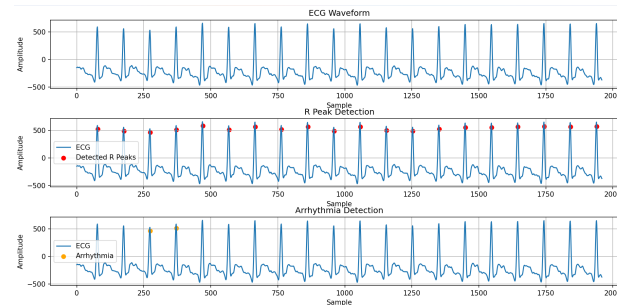


Fig. 5. Graph

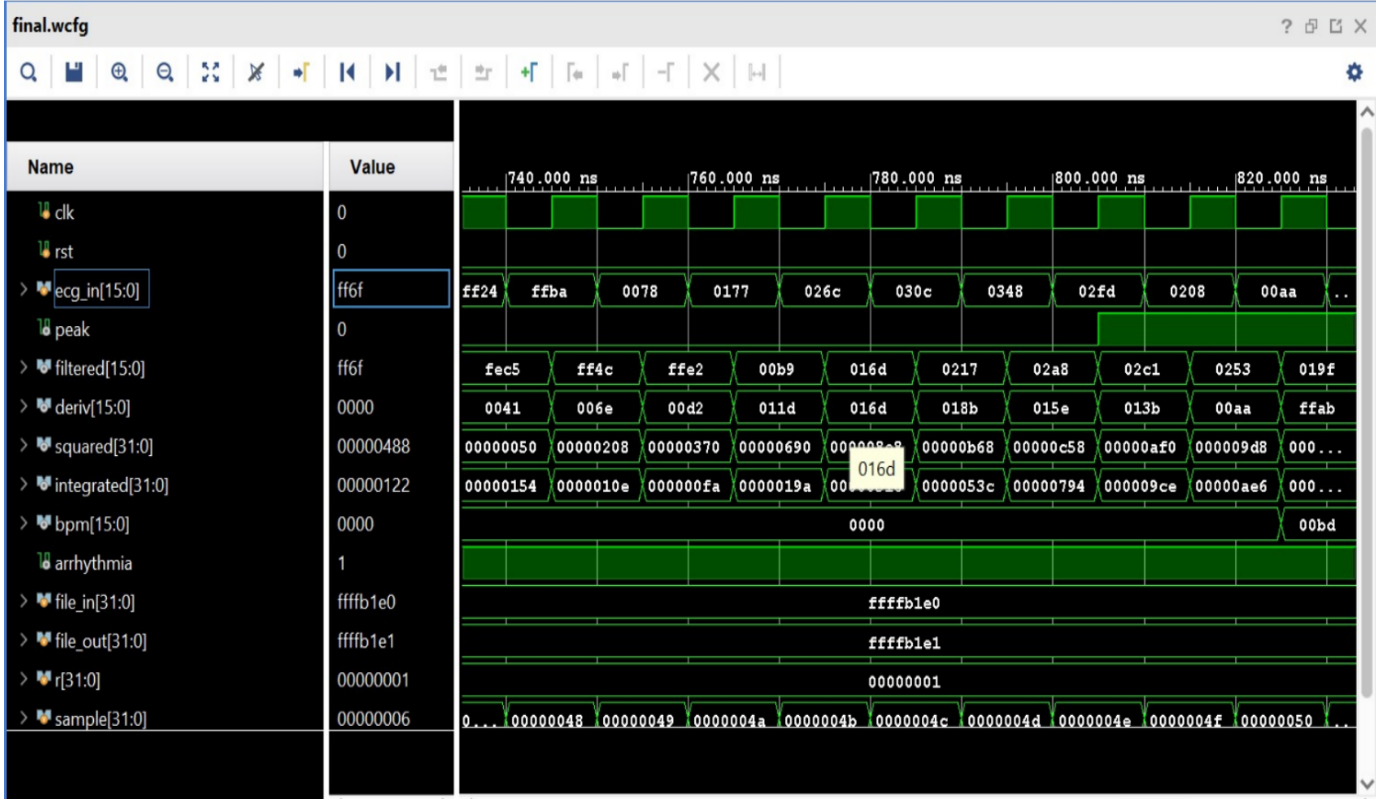


Fig. 6. Waveform

TABLE I
PERFORMANCE COMPARISON BETWEEN EXISTING AND PROPOSED
FPGA-BASED ECG SYSTEM

Metric	Existing System	Proposed FPGA System
Execution Time	High	Very Low
Latency	High	Low
Throughput	Low	High
Hardware Utilization	Not Applicable	Efficient
Power Efficiency	Low	High
Real-Time Processing	No	Yes
Detection Speed	Slow	Fast
System Complexity	Low	Moderate
Accuracy	85–92%	95–98%

TABLE II
FPGA RESOURCE UTILIZATION OF PROPOSED ECG SYSTEM

Resource	Used	Available	Utilization (%)
Slice LUTs	142	32600	0.44
Slice Registers	247	65200	0.38
DSP Blocks	1	120	0.83
Block RAM	0	75	0.00
I/O Pins	122	210	58.10

The FPGA implementation of the proposed ECG processing system demonstrates highly efficient resource utilization, with only 0.44% of LUTs and 0.38% of registers utilized. This indicates that the design is lightweight and scalable for larger implementations.

Timing analysis shows zero negative slack and no failing

endpoints, confirming that the design meets timing requirements. However, no explicit timing constraints were defined, which should be addressed in future work for accurate timing validation.

Power analysis reveals that the total on-chip power consumption is relatively high at 85.249 W, with 99% attributed to dynamic power. A significant portion (92%) of this power is consumed by I/O operations, indicating that external data transfer is the primary contributor to power usage. The junction temperature reaches 125°C, suggesting a need for power optimization and thermal management in practical deployments.

VI. CONCLUSION

This work presents an efficient FPGA-based implementation of an ECG signal processing system for real-time arrhythmia detection. The proposed design follows a structured signal processing pipeline consisting of filtering, differentiation, squaring, and moving window integration to accurately identify QRS complexes and detect abnormalities in cardiac rhythms.

The implementation results demonstrate that the design is highly resource-efficient, utilizing only 0.44% of LUTs and 0.38% of registers, which confirms its suitability for low-cost and scalable hardware deployment. Timing analysis indicates zero negative slack and no failing endpoints, ensuring reliable operation under the given conditions. Functional verification through simulation waveforms confirms correct signal trans-

formation across all processing stages, including successful peak detection and arrhythmia indication.

However, power analysis reveals that the total on-chip power consumption is relatively high, primarily due to I/O switching activity, contributing to 92% of the dynamic power. This also results in an elevated junction temperature, highlighting the need for optimization in I/O handling and power management for practical implementations.

Overall, the proposed system achieves high-speed, real-time ECG signal processing with minimal hardware utilization, making it a promising solution for portable and embedded healthcare applications. Future work will focus on optimizing power consumption, applying timing constraints for improved accuracy, and integrating real-time sensor interfaces for deployment in wearable medical devices.

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